DIPHENHYDRAMINE

Strongly sedating

Avoid in elderly

CHEMICAL NAME 2-(diphenylmethoxy)-N,N-dimethylethanamine (hydrochloride)
CLASS First-generation (sedating) H1 antihistamine (ethanolamine)
TRADE NAMES Benadryl · Nytol · Sominex · Allerdryl

THERAPEUTIC USES / INDICATIONS

- Allergic reactions, urticaria, pruritus.
- Insomnia (short-term).
- Motion sickness.
- Acute dystonic reactions from antipsychotics/metoclopramide (IV/IM); cough (syrups).

ADMINISTRATION

- **PO** tablets, capsules, syrup.
- **IV** slowly (max 25 mg/min) or **IM** deep.

DOSAGES

- **Adult allergy:** 25–50 mg every 4–6 h; max 300 mg/day.
- **Insomnia:** 50 mg 20–30 min before bed.
- **Dystonic reaction:** 25–50 mg IV/IM.
- **Children 6–12 yr:** 12.5–25 mg every 4–6 h (avoid <6 yr for OTC use).

PHARMACOKINETICS

- **Onset:** PO 15–60 min; IV immediate. **Duration:** 4–6 h.
- Metabolised in liver (CYP2D6 inhibitor).
- Excreted in urine. **t½:** 3–9 h (longer in elderly).

PHARMACODYNAMICS

- **H1 antagonist** with marked antimuscarinic and sedative effects.
- Central anticholinergic action relieves dystonia; antiemetic effect.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Marked drowsiness, dizziness.
- Dry mouth, constipation, blurred vision.
- Thickened bronchial secretions.

ADVERSE EFFECTS (SERIOUS)

- Confusion, delirium in elderly (Beers list).
- Urinary retention.
- Paradoxical excitation in children.
- Overdose: seizures, arrhythmias, anticholinergic toxicity.

CONTRAINDICATIONS & CAUTIONS

- Neonates; breastfeeding.
- MAOI use.
- **Caution:** glaucoma, BPH, asthma attack, elderly.

KEY INTERACTIONS

- **CNS depressants, alcohol:** ↑ sedation.
- **Metoprolol and other CYP2D6 substrates:** ↑ levels.
- **Other anticholinergics:** additive.

NURSING MANAGEMENT

- Assess allergic symptoms or dystonia response.
- Fall precautions; bed rails for elderly.
- Monitor urine output.
- Avoid as sleep aid in elderly.

PATIENT EDUCATION

- Causes drowsiness; no driving or alcohol.
- Do not combine with other cold or sleep products.
- For short-term use only.

KEY POINTS

- Sedating H1 antihistamine with strong anticholinergic action.
- Avoid in elderly (delirium).
- Treats acute dystonia.
- No alcohol or driving.

SOURCES CHECKED **[477]** 21 CFR Part 341 (US OTC monograph) **[478]** Diphenhydramine Hydrochloride, IM, IV (US label) **[479]** Diphenhydramine Hydrochloride, oral (US label)
Full references p. 432–457